

HDOC Day-15

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SUBJECT: C programming

DATE: 8-09-26

Question 1: Check Whether All Digits Are Even

Aim: To write a switch-case based C program that accepts a positive integer and checks whether every digit of the number is even.

1. Algorithm

1. Start the program.
2. Display a menu: 1 to check all digits are even, 2 to exit.
3. Read the user choice.
4. Use switch-case to process the choice.
5. For choice 1, read a positive integer.
6. Copy the number into a temporary variable.
7. Repeat while the temporary value is greater than 0: extract the last digit using temp % 10.
8. If the digit is odd, set the flag to false and stop checking; otherwise remove the last digit using temp / 10.
9. If the flag remains true, print "All digits are even"; otherwise print "All digits are not even".
10. Handle exit and invalid menu choices, then stop.

2. Test Case Table

Test Case	Choice	Input Number	Expected Output
TC1	1	12345	All digits are not even
TC2	1	24680	All digits are even
TC3	1	8642	All digits are even
TC4	1	24861	All digits are not even

3. C Program

```
#include <stdio.h>
```

```

int main() {
    int choice, number, digit, temp, allEven = 1;

    printf("1. Check whether all digits are even\n");
    printf("2. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            printf("Enter a positive integer: ");
            scanf("%d", &number);

            if (number <= 0) {
                printf("Please enter a positive integer.\n");
                break;
            }

            temp = number;
            while (temp > 0) {
                digit = temp % 10;
                if (digit % 2 != 0) {
                    allEven = 0;
                    break;
                }
                temp = temp / 10;
            }

            if (allEven)
                printf("All digits are even\n");
            else
                printf("All digits are not even\n");
            break;

        case 2:
            printf("Program ended.\n");
            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}

```

4. Compile the Program

Save the program and compile it using GCC:

```
gcc alldigitseven.c -lm
Compilation status: Successful (no errors).
```

5. Execute the Program

Run the compiled program and enter the required choice and number:

```
./a.out
```

```

(kali㉿kali)-[~/Desktop/cprog]
$ gcc alldigitseven.c -lm

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
1. Check whether all digits are even
2. Exist
Enter your choice: 1
Enter a positive integer: 12345
All digitd are not even

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
1. Check whether all digits are even
2. Exist
Enter your choice: 1
Enter a positive integer: 24680
All digits are even

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
1. Check whether all digits are even
2. Exist
Enter your choice: 1
Enter a positive integer: 8642
All digits are even

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
1. Check whether all digits are even
2. Exist
Enter your choice: 1
Enter a positive integer: 24861
All digitd are not even

```

6. Test Result

Test Case	Expected Result	Actual Result	Status
TC1	All digits are not even	All digits are not even	PASS
TC2	All digits are even	All digits are even	PASS
TC3	All digits are even	All digits are even	PASS
TC4	All digits are not even	All digits are not even	PASS

Question 2: Check Whether All Digits Are Prime

Aim: To write a switch-case based C program that accepts a positive integer and checks whether every digit is a prime digit (2, 3, 5, or 7).

1. Algorithm

1. Start the program.
2. Display a menu: 1 to check all digits are prime, 2 to exit.
3. Read the user choice.
4. Use switch-case to process the choice.
5. For choice 1, read a positive integer.
6. Copy the number into a temporary variable.
7. Repeat while the temporary value is greater than 0: extract the last digit using $\text{temp} \% 10$.
8. Check whether the digit is 2, 3, 5, or 7. If not, set the flag to false and stop checking.
9. Remove the checked digit using $\text{temp} / 10$ and continue.
10. If the flag remains true, print "All digits are prime"; otherwise print "All digits are not prime".
11. Handle exit and invalid menu choices, then stop.

2. Test Case Table

Test Case	Choice	Input Number	Expected Output
TC1	1	2357	All digits are prime
TC2	1	2379	All digits are not prime
TC3	1	7523	All digits are prime
TC4	1	2057	All digits are not prime

3. C Program

```
#include <stdio.h>

int main() {
    int choice, number, digit, temp, allPrime = 1;
```

```
printf("1. Check whether all digits are prime\n");
printf("2. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        printf("Enter a positive integer: ");
        scanf("%d", &number);

        if (number <= 0) {
            printf("Please enter a positive integer.\n");
            break;
        }

        temp = number;
        while (temp > 0) {
            digit = temp % 10;
            if (digit != 2 && digit != 3 && digit != 5 && digit != 7) {
                allPrime = 0;
                break;
            }
            temp = temp / 10;
        }

        if (allPrime)
            printf("All digits are prime\n");
        else
            printf("All digits are not prime\n");
        break;

    case 2:
        printf("Program ended.\n");
        break;

    default:
        printf("Invalid choice\n");
}

return 0;
}
```

4. Compile the Program

Save the program and compile it using GCC:

```
gcc alldigitsprime.c -lm
Compilation status: Successful (no errors).
```

5. Execute the Program

Run the compiled program and enter the required choice and number:

```
./a.out
```

```

(kali㉿kali)-[~/Desktop/cprog]
$ gcc alldigitsprime.c -lm

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
1. Check whether all digits are prime
2. Exit
Enter your choice: 1
Enter a psoitive integer: 2357
All digits are prime

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
1. Check whether all digits are prime
2. Exit
Enter your choice: 1
Enter a psoitive integer: 2379
all digits are not prime

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
1. Check whether all digits are prime
2. Exit
Enter your choice: 1
Enter a psoitive integer: 7523
All digits are prime

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
1. Check whether all digits are prime
2. Exit
Enter your choice: 1
Enter a psoitive integer: 2057
all digits are not prime

```

6. Test Result

Test Case	Expected Result	Actual Result	Status
TC1	All digits are prime	All digits are prime	PASS
TC2	All digits are not prime	All digits are not prime	PASS
TC3	All digits are prime	All digits are prime	PASS
TC4	All digits are not prime	All digits are not prime	PASS